

2. Phototransistor Sensor

BPW40/BPW77 orientation

Check the exact datasheet for your part.

Collector -> sensor node / GPIO34

Emitter -> GND

Base (if present) -> not connected

Bias network

3V3 -> 1 Mohm resistor -> GPIO34 sensor node

Phototransistor collector -> same GPIO34 node

Phototransistor emitter -> GND

Firmware

Analog input: GPIO34

Alarm threshold is adjustable in Home Assistant.